Equivalent expressions — Part 1

Advanced Math · Math · 58 questions

1.1 04639-DC · Easy

Which of the following is equivalent to $3(x+5)-6$?

A. $3x-3$

B. $3x-1$

C. $3x+9$

D. $15x-6$

1.2 be5faa6c-0169-41dd-90d9-1eb1a8bdeca1 · Easy

Which expression is equivalent to $256w^2 - 676$?

A. $(16w - 26)(16w - 26)$

B. $8w - 138w + 13$

C. $(8w - 13)(8w - 13)$

D. $16w - 2616w + 26$

1.3 21bb428c-50de-4b82-ba02-72c7b6db2d11 · Easy

Which expression is a factor of $2x^2 + 38x + 10$?

A. 2

B. $5x$

C. $38x$

D. $2x^2$

1.4 440a7343-d56b-4b84-936d-ec6c5596d326 · Easy

Which expression is equivalent to $8yzy7z$?

A. $56y^2z^2$

B. $56y^2z$

C. $56yz$

D. $16yz$

1.5 a81bca68-7fa7-4265-906c-6d5f61769b51 · Easy

Which expression is equivalent to $d - 68d^2 - 3$?

A. $8d^3 - 14d^2 - 3d + 18$

B. $8d^3 - 17d^2 + 48$

C. $8d^3 - 48d^2 - 3d + 18$

D. $8d^3 - 51d^2 + 48$

1.6 14779160-e23c-403b-98af-36fdefea4b8e · Easy

Which expression is equivalent to $100x^2 + 2{,}600$?

A. $10026x^2$

B. $100x^2 + 26$

C. $2{,}600100x^2$

D. $2{,}600100x^2 + 1$

1.7 f0feb4ae-ee24-4b97-b265-104550acbce4 · Easy

Which expression is equivalent to $x^2 + 3x - 40$?

A. $x - 4x + 10$

B. $x - 5x + 8$

C. $x - 8x + 5$

D. $x - 10x + 4$

1.8 52214fdf-56c8-448c-ac5e-c62e616b3288 · Easy

Which expression is equivalent to $34x + 34y$?

A. $34xy$

B. $34(x + y)$

C. $68y$

D. $68x$

1.9 029332-DC · Easy

Which of the following expressions is equivalent to $2a^2(a+3)$?

A. $5a^3$

B. $8a^5$

C. $2a^3+3$

D. $2a^3+6a^2$

1.10 1620a4b4-e1d2-49e7-aae0-096ea9b959f1 · Easy

Which expression is equivalent to $4(x^2 + 6)$?

A. $4x^2 + 24$

B. $4x^2 + 10$

C. $4x^2 + 6$

D. $4x^2 - 2$

1.11 016559-DC · Easy

Which of the following is equivalent to $2(x^2-x)+3(x^2-x)$?

A. $5x^2-5x$

B. $5x^2+5x$

C. 5x

D. $5x^2$

1.12 4aae6d4f-125f-4c81-812c-7d232ae7986c · Easy

Which expression is equivalent to $13x^2 - 7x^2$?

A. $-91x^2$

B. $6x^2$

C. $20x^2$

D. $40x^2$

1.13 022215-DC · Easy

The expression $2x^2+ax$ is equivalent to $x(2x+7)$ for some constant a. What is the value of a ?

A. 2

B. 3

C. 4

D. 7

1.14 05476-DC · Easy

Which expression is equivalent to $(2x^2-4)-(-3x^2+2x-7)$?

A. $5x^2-2x+3$

B. $5x^2+2x-3$

C. $-x^2-2x-11$

D. $-x^2+2x-11$

1.15 9c3d170a-52ba-4266-b573-bfa73fbbe581 · Easy

Which expression is equivalent to $9x^2+7x^2+9x$?

A. $63x^4+9x$

B. $9x^2+16x$

C. $25x^5$

D. $16x^2+9x$

1.16 82d332a7-36d9-4422-9783-754cdaf665a2 · Easy

Which expression is equivalent to $16x^3y^2 + 14xy$?

A. $2xy8xy + 7$

B. $2xy8x^2y + 7$

C. $14xy2x^2y + 1$

D. $14xy8x^2y + 1$

1.17 11aa8904-6d75-4f09-b150-a48f09955868 · Easy

Which expression is equivalent to $9x^3 + 5x + 7 + 6x^3 + 5x^2 - 5$?

A. $15x^6 + 5x^2 - 5x - 35$

B. $15x^3 + 10x^2 + 2$

C. $15x^6 + 5x^2 + 5x + 2$

D. $15x^3 + 5x^2 + 5x + 2$

1.18 8a43ff56-7c88-420c-8b4c-bc730b55e931 · Easy

Which expression is equivalent to $9x^2 + 6 - 8x^2$?

A. 7

B. $17x^2 + 6$

C. $x^2 - 6$

D. $x^2 + 6$

1.19 4331067c-b0f9-4aaf-a21b-3d660144cf68 · Easy

Which expression is equivalent to $8 + d^2 + 3$?

A. $d^2 + 24$

B. $d^2 + 11$

C. $d^2 + 5$

D. $d^2 - 11$

1.20 031616-DC · Easy

Which of the following is equivalent to $2x^3 + 4$?

A. $4(x^3 + 4)$

B. $4(x^3 + 2)$

C. $2(x^3 + 4)$

D. $2(x^3 + 2)$

1.21 027693-DC · Easy

$5x + 15$

Which of the following is equivalent to the given expression?

A. $5(x + 3)$

B. $5(x + 10)$

C. $5(x + 15)$

D. $5(x + 20)$

1.22 4363efa2-3674-407a-a249-7f0cf0daff53 · Easy

Which expression is equivalent to $5x^2 - 50xy^2$?

A. $5xx - 10y^2$

B. $5xx - 50y^2$

C. $5x^2\,10xy^2$

D. $5x^2\,50xy^2$

1.23 885afb10-2178-4564-84ab-6e185b004a41 · Easy

Which expression is equivalent to $20w - 4w + 3w$?

A. $10w$

B. $13w$

C. $19w$

D. $21w$

1.24 a7c54228-4ce0-4f08-8ae4-811ea92339af · Easy

Which expression is equivalent to $23x^3 + 2x^2 + 9x$?

A. $23xx^2 + 2x + 9$

B. $9x23x^3 + 2x^2 + 1$

C. $x23x^2 + 2x + 9$

D. $34x^3 + x^2 + x$

1.25 017096-DC · Easy

Which of the following expressions is equivalent to the sum of (r^3+5r^2+7) and $(r^2+8r+12)$?

A. r^5+13r^3+19

B. $2r^3+13r^2+19$

C. $r^3+5r^2+7r+12$

D. $r^3+6r^2+8r+19$

1.26 70a9ddd5-661b-45e3-b279-9c6de073b79d · Easy

Which expression is equivalent to $19x^2 - 7$?

A. $19x^2 - 133$

B. $19x^2 - 26$

C. $19x^2 - 7$

D. $19x^2 + 12$

1.27 707e44df-b9c2-476c-a030-957fe31a519a · Easy

Which expression is equivalent to $50x^2 + 5x^2$?

A. $250x^2$

B. $10x^2$

C. $45x^2$

D. $55x^2$

1.28 56e878c9-7295-480c-8ba1-5c2246df36be · Easy

Which expression is equivalent to $15w^2 + 8w$?

A. $w15w + 8$

B. $8w15w + 1$

C. $15w^2\,8w + 1$

D. $23w^2 + w$

1.29 5d93e88d-9666-4e29-baab-4dcd15423597 · Easy

Which expression is equivalent to $11x^3 - 5x^3$?

A. $16x^3$

B. $6x^3$

C. $6x^6$

D. $16x^6$

1.30 238bf570-e789-4f6b-9588-2ff4572b2368 · Easy

Which expression is equivalent to $6x + 5x + 4y$?

A. $15x$

B. $15y$

C. $11x + 4y$

D. $30x + 4y$

1.31 e31accbd-71ed-4323-82c0-b30186a84861 · Easy

Which expression is equivalent to $16x + 15$?

A. $16x + 31$

B. $16x + 240$

C. $16x + 1$

D. $16x + 15$

1.32 e2375cf4-517a-4d38-94ef-5c6c3cc7b879 · Easy

Which expression is equivalent to $5x^5 - 6x^4 + 8x^3$?

A. $x^4\,5x - 6$

B. $x^3\,5x^2 - 6x + 8$

C. $8x^3\,5x^2 - 6x + 1$

D. $6x^5 - 6x^4 + 8x^3 + 1$

1.33 8ef4c8a0-31ce-4c6e-b0e5-d7b0b944b67d · Easy

Which expression is equivalent to $9x^2 + 5x$?

A. $x9x + 5$

B. $5x9x + 1$

C. $9xx + 5$

D. $x^2\,9x + 5$

1.34 295f9556-4a27-49ca-b269-fb8917b7e050 · Easy

Which expression is equivalent to $12x + 27$?

A. $129x + 1$

B. $2712x + 1$

C. $34x + 9$

D. $39x + 24$

1.35 ba751bc0-12b0-41ac-9d53-b1375e5b11f4 · Easy

Which expression is equivalent to $64t^2s^3 - 56t^3s$?

A. $4ts16s^2 - 14ts$

B. $4ts16t^2s^2 - 14t$

C. $4t^2s16ts - 14s$

D. $4t^2s16s^2 - 14t$

1.36 025717-DC · Easy

$(2x^3 + 3x)(x^3 - 2x)$

Which of the following is equivalent to the expression above?

A. $x^3 + 5x$

B. $3x^3 + x$

C. $2x^6 - x^4 - 6x^2$

D. $3x^6 - x^4 - 6x^2$

1.37 79ab98cb-39f9-474f-8017-4a2677075b32 · Easy

Which expression is equivalent to $17x^2 - 100y^2$?

A. $17x - 2yx - 50y$

B. $17x - 2yx + 50y$

C. $17x - 10yx - 10y$

D. $17x - 10yx + 10y$

1.38 027123-DC · Easy

$(x+5)+(2x-3)$

Which of the following is equivalent to the given expression?

A. $3x-2$

B. $3x+2$

C. $3x-8$

D. $3x+8$

1.39 7a034397-6081-46d4-b5f6-0a2735b3a604 · Easy

Which expression is equivalent to $9x + 6x + 2y + 3y$?

A. $3x + 5y$

B. $6x + 8y$

C. $12x + 8y$

D. $15x + 5y$

1.40 8dbb28ea-e265-4e97-95ca-ce81e767f4e8 · Easy

Which expression is equivalent to $m^4 q^4 z^{-1} mq^5 z^3$, where m, q, and z are positive?

A. $m^4 q^{20} z^{-3}$

B. $m^5 q^9 z^2$

C. $m^6 q^8 z^{-1}$

D. $m^{20} q^{12} z^{-2}$

1.41 028884-DC · Easy

Which of the following expressions is equivalent to $2(ab-3)+2$?

A. $2ab-1$

B. $2ab-4$

C. $2ab-5$

D. $2ab-8$

1.42 ff1e479f-7df1-4306-a31d-d9354d96102e · Easy

Which expression is equivalent to $12x^3 - 5x^3$?

A. $7x^6$

B. $17x^3$

C. $7x^3$

D. $17x^6$

1.43 d3f7c3fc-9ac9-456f-8bf8-53653016b0fb · Easy

Which expression is equivalent to $2x^2 + x - 9 + x^2 + 6x + 1$?

A. $2x^2 + 7x + 10$

B. $2x^2 + 6x - 8$

C. $3x^2 + 7x - 10$

D. $3x^2 + 7x - 8$

1.44 025800-DC · Medium

Which of the following is equivalent to $\sqrt[4]{x^2+8x+16}$, where $x > 0$?

A. $(x+4)^4$

B. $(x+4)^2$

C. $(x+4)$

D. $(x+4)^{\frac{1}{2}}$

1.45 d469dd2b-510b-49e3-b521-1271ecbbfa1e · Medium

Which expression is equivalent to $\frac{h^{15}q^7}{h^5q^{21}}$, where $h > 0$ and $q > 0$?

A. $\frac{h^{10}}{q^{14}}$

B. $\frac{h^3}{q^3}$

C. $h^{10}q^{14}$

D. h^3q^3

1.46 afeb4dfe-3156-4234-af28-242e9b1500f8 · Medium

$$(5x^3 - 3) - (-4x^3 + 8)$$

The given expression is equivalent to $bx^3 - 11$, where b is a constant. What is the value of b?

Student-produced response: _____

1.47 f4365bf0-90a1-4025-b4b6-b2ba9e88b93a · Medium

If $a = 4k + 5r$ and $b = 7k - 12r + 3$, which expression is equivalent to $a - b$?

A. $-3k + 17r + 3$

B. $-3k + 17r - 3$

C. $-3k - 7r - 3$

D. $-3k - 7r + 3$

1.48 09404-DC · Medium

Which of the following is equivalent to $(1-p)(1+p+p^2+p^3+p^4+p^5+p^6)$?

A. $1-p^8$

B. $1-p^7$

C. $1-p^6$

D. $1-p^5$

1.49 34c2743d-443b-43f9-ad72-c43601418315 · Medium

Which expression is equivalent to $\sqrt[9]{x^9 y^9}$, where x and y are positive?

A. $xy^{\frac{7}{9}}$

B. $xy^{\frac{9}{7}}$

C. xy^{16}

D. xy^{63}

1.50 63296c8a-e373-4a99-89f1-f1ceaa55e479 · Medium

The expression $\frac{24}{6x+42}$ is equivalent to $\frac{4}{x+b}$, where b is a constant and $x > 0$. What is the value of b?

A. 7

B. 10

C. 24

D. 252

1.51 027180-DC · Medium

$2x^2+5x-12$

If the given expression is rewritten in the form $(2x-3)(x+k)$, where k is a constant, what is the value of k ?

Student-produced response: _____

1.52 ba562150-ace0-4dca-b1c3-c7c1019ccee7 · Medium

$$f(x) = x^2 + bx$$

$$g(x) = 9x^2 - 27x$$

Functions f and g are given, and in function f, b is a constant. If $f(x) \cdot g(x) = 9x^4 - 26x^3 - 3x^2$, what is the value of b?

A. -26

B. $-\frac{26}{9}$

C. $\frac{1}{9}$

D. 9

1.53 322bacf8-a04d-4c31-8a29-228d7b8ccca3 · Medium

Which expression is equivalent to $6x^8y^2 + 12x^2y^2$?

A. $6x^2y^2 2x^6$

B. $6x^2y^2x^4$

C. $6x^2y^2x^6 + 2$

D. $6x^2y^2x^4 + 2$

1.54 025483-DC · Medium

$(2x+5)^2-(x-2)+2(x+3)$

Which of the following is equivalent to the expression above?

A. $4x^2+21x+33$

B. $4x^2+21x+29$

C. $4x^2+x+29$

D. $4x^2+x+33$

1.55 4544da8e-b007-4229-a855-ce10c2bbf557 · Medium

Which expression is equivalent to $a^{\frac{11}{12}}$, where $a > 0$?

A. $\sqrt[12]{a^{132}}$

B. $\sqrt[144]{a^{132}}$

C. $\sqrt[12]{a^{132}}$

D. $\sqrt[11]{a^{132}}$

1.56 2f0bba12-b9f1-460c-b120-0adaf38cf4db · Medium

Which expression is equivalent to $x^{\frac{1}{14}}$, where $x > 0$?

A. $\frac{1}{14} \cdot x$

B. $\sqrt[14]{x}$

C. $14 \cdot x$

D. x^{14}

1.57 026581-DC · Medium

If $x \neq 0$, which of the following expressions is equivalent to $\frac{\sqrt{16x^4y^8}}{x^3}$?

A. $8x^2y^4$

B. $4xy^4$

C. $4x^{-2}y^2$

D. $4x^{-1}y^4$

1.58 37e39587-24a9-4c5a-b541-d42a0a10ed7a · Medium

Which expression is equivalent to $x^2 + 11^2 + x - 5x + 5$?

A. $x^4 + 23x^2 - 14$

B. $x^4 + 23x^2 + 96$

C. $x^4 + 12x^2 + 121$

D. $x^4 + x^2 + 146$